Vocabulary: Explainable AI – Week 01
XAI Foundations & Interpretable Models

Kristian Boroz

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Explainability & Model Interpretation

- **Explainable AI (XAI)**: Methods that help people understand model predictions. The slides describe explanations as useful for debugging, trust, fairness, and compliance.
- **Interpretability**: How easily a person can understand a model or its predictions. Linear models and small decision trees are examples of interpretable models discussed in the slides.
- **Intrinsic Explanation**: An explanation built into the model itself. Examples include linear weights and decision-tree rules.
- **Post-hoc Explanation**: An explanation created after a model has been trained. Examples in the slides include SHAP, LIME, and PDP.
- **Global Explanation**: Describes overall model behavior. The slides give permutation feature importance as an example.
- **Local Explanation**: Explains a specific prediction, such as why a loan application was rejected. Counterfactuals and LIME are examples.
- **Model-specific Method**: Uses details of a model's internal structure, such as weights or gradients. Examples include Grad-CAM and TreeSHAP.
- **Model-agnostic Method**: Uses only model inputs and outputs rather than its internal structure. Examples include PFI and KernelSHAP.
- **Saliency Map**: A visual explanation showing which pixels support an image prediction. The slides note that different methods, or harmless changes to an image, can produce different maps.

Linear and Logistic Models

- **Feature**: An input variable used by a model to make a prediction. The slides use x_j to denote a feature.
- **Linear Regression**: A model whose prediction is a weighted sum of its features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

  Here, β_j is the change in $\hat{y}$ for a one-unit increase in x_j, with all other features fixed.

- **Logistic Regression**: A model that expresses the log-odds of $y = 1$ as a linear combination of features:

$$\log \frac{P(y = 1 \mid x)}{1 - P(y = 1 \mid x)} = \beta_0 + \sum_j \beta_j x_j$$

- **Coefficient**: A value β_j in a linear or logistic model. The slides caution that coefficients can be difficult to compare across features with different scales, and can become unstable when features are correlated.

- **Odds Ratio**: In the logistic regression formula, the odds ratio associated with coefficient β_j is:

$$e^{\beta_j}$$

  The slides show calculating odds ratios by exponentiating model coefficients.

- **Linearity Assumption**: In a linear model, the effect of a feature is assumed to remain constant across its range.

Decision Trees & Model Pitfalls

- **Decision Tree**: A model that makes predictions by following a path of if–then rules. The slides note that trees can handle non-linearity and interactions.

- **Tree Depth**: The number of decision steps along a tree path. The slides describe shallow trees as more readable, while deep trees can be harder to interpret.

- **Shortcut**: An unintended cue a model uses to make a prediction. In the slides' husky-versus-wolf example, the classifier relied on snow in the background.

- **Confounder**: A factor that can lead a model to learn a misleading relationship. In the slides' pneumonia example, asthma appeared to lower risk because asthma patients received intensive care.

- **Data Leakage**: A problem to look for when debugging a model, listed alongside shortcuts in the slides.

- **Tree Instability**: Small changes in training data can change a decision tree substantially.

- **Interpretability–Accuracy Tradeoff**: The slides describe a tendency for more complex models to be less interpretable, while often being more accurate on complex data. They recommend fitting an interpretable baseline first.

Common XAI Shortnames

Shortname	Meaning
XAI	Explainable Artificial Intelligence
SHAP	Method named in the slides as an example of post-hoc explanation
LIME	Method named in the slides as an example of post-hoc, local explanation
PDP	Partial Dependence Plot; named in the slides as a post-hoc method
PFI	Permutation Feature Importance; named as a global, model-agnostic method
ICE	Individual Conditional Expectation; listed with PDP in the course map
ALE	Accumulated Local Effects; listed in the course map
IG	Integrated Gradients; listed as a model-specific vision method
LRP	Layer-wise Relevance Propagation; listed as a model-specific vision method
Grad-CAM	Model-specific vision explanation method listed in the slides
DiCE	Method listed with counterfactual explanations